Total No. of Questions : 4]

SEAT No. :

PE-316

[Total No. of Pages : 2

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B.E. (Electronics and Computer Engineering) (Insem.)

INFORMATION AND CYBER SECURITY

(2019 Pattern) (Semester - VII) (Elective - III) (410344C)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q1 or Q2 and Q3 or Q4.***
2) ***Figures to the right side indicate full marks.***
3) ***Assume suitable data, if necessary.***

Q1) a) Explain CIA triad with neat diagram **[5]**

b) What is Information security? List the elements of Information security **[5]**

c) Define the goals of Security & Specify mechanisms to achieve each goal **[5]**

OR

Q2) a) Write short note on **[5]**

i) Security Threats with suitable example

ii) Information Assurance

b) What is meant by operational Security? Why it is important? **[5]**

c) List & Explain Basic Components of Information Security. **[5]**

Q3) a) Use play fair cipher to encrypt following message "This is columnar transposition" use key APPLE **[5]**

b) Explain the operation in key expansion process in AES algorithm. **[5]**

c) What is the difference between a monoalphabetic cipher and a polyalphabetic cipher? Explain with example. **[5]**

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OR

Q4) a) Explain with diagram DES Encryption Algorithm. Explain Advantages and Drawbacks of DES. **[5]**

b) Differentiate Between Substitution Cipher and Transposition Cipher. **[5]**

c) Using extended Euclidean algorithm, find the multiplicative inverse of

i) 1234 mod 4321

ii) 24140 mod 40902

[5]

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